

MARKETING CAMPAIGN ANALYTICS PROJECT REPORT

*A comprehensive analysis of customer behaviour, spending patterns,
campaign effectiveness, and actionable recommendations*

Domain: Marketing / Customer Analytics
Tools: Python | SQL | EDA | Streamlit

1. Problem Understanding

1.1 Business Context

A retail company has run multiple marketing campaigns and recorded detailed information about customers, their demographics, spending across product categories, channel-wise purchases, and whether they accepted specific campaigns. Management requires a consolidated analytics solution to understand who its best customers are, what drives campaign acceptance, and how different customer segments behave across products and channels.

1.2 Dataset Overview

The dataset contains 55,986 customer records with 36 features spanning demographics, spending, channel usage, and campaign responses.

Category	Key Variables
Demographics	ID, Year_Birth, Education, Marital_Status, Income, Kidhome, Teenhome, Country
Customer Relationship	Dt_Customer, Recency, Tenure_days
Spending (2 years)	MntWines, MntFruits, MntMeatProducts, MntFishProducts, MntSweetProducts, MntGoldProds
Channel Usage	NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumStorePurchases, NumWebVisitsMonth
Campaigns	AcceptedCmp1, AcceptedCmp2, AcceptedCmp3, AcceptedCmp4, AcceptedCmp5, Response
Other	Complain, Total_Amt, Total_Purchases, Age, Age_Group, Income_Group, Childrens

2. Approach & Methodology

2.1 Data Cleaning (Python)

- Loaded the raw marketing_data.csv using Pandas.
- Converted Dt_Customer to datetime and derived Tenure_days from the reference date.
- Calculated Age from Year_Birth (reference year 2014).
- Derived Total_Amt as the sum of all Mnt* spending fields.
- Derived Total_Purchases as the sum of all channel purchase counts.
- Created Childrens = Kidhome + Teenhome.
- Handled missing values in Income by dropping null rows (24 records removed).
- Removed outliers: customers with unrealistic ages (>100) and extreme income values.
- Encoded Age_Group (Young, Middle Aged, Senior) and Income_Group (Low to High) using pd.cut().

- Exported cleaned dataset as `Cleaned_marketing_data.csv` with 55,986 rows and 36 columns.

2.2 Exploratory Data Analysis (Python)

- Univariate analysis: distributions of Age, Income, Total_Amt, Recency, and web visits.
- Bivariate analysis: Response vs. Income, Age, Education, and product spend.
- Multivariate analysis: heatmaps of Education x Income Group vs. average spend and response rate.
- Segment-level comparisons: channel usage by age group, spend by country and marital status.

2.3 Rule-Based Segmentation

Six business-defined segments were implemented using if-else logic and `pd.cut()` in Python and CASE statements in SQL:

Segment	Rule	% of Customers
High Income	Income > \$75,000	~22%
Young Customer	Age < 30	~8%
Campaign Responder	Response = 1	~15%
High Web Engaged	NumWebVisitsMonth > 5	~38%
Family Customer	Childrens > 0	~64%
High Spender	Total_Amt > 90th percentile (~\$1,200)	~10%

2.4 SQL Data Modelling

- Created a MySQL database (market) with a main table `market_data`.
- Loaded the cleaned CSV into the table using Python (SQLAlchemy / mysql-connector).
- Wrote 15 analytical queries covering KPIs, segment summaries, channel analysis, and campaign performance.

2.5 Dashboard (Streamlit)

- Built a multi-tab interactive dashboard using Streamlit and Plotly.
- Sidebar filters: Country, Education, Age Group, Income Group, Marital Status.
- Five tabs: Campaign Performance, Customer Segments, Spending Patterns, Channel Analysis, Top Customers.

3. Key Findings

3.1 Overall KPIs

Metric	Value
Total Customers	55,986
Overall Response Rate	~15%
Average Customer Spend	\$606
Total Revenue	\$33.9M
Average Customer Income	\$52,247
Complaint Rate	~1%

3.2 Campaign Performance

- Campaign 4 and Campaign 5 had the highest acceptance rates among all five campaigns.
- Campaign 2 had the lowest acceptance rate, suggesting poor targeting or messaging.
- Customers who accepted 2 or more campaigns spent significantly more on average than those who accepted none.
- Responders spent nearly 3x more than non-responders across all product categories, especially Wine and Meat.

3.3 Customer Demographics

- The majority of customers fall in the Middle Aged (30–59) and Senior (60+) age groups.
- Graduation is the most common education level (~50%), followed by PhD and Master.
- Spain accounts for the largest share of customers; Australia and Canada follow.
- 64% of customers have at least one child at home, making Family Customer the largest segment.

3.4 Spending Patterns

- Wine is the top-selling product category by both average spend and total revenue.
- Meat products rank second, followed by Gold products.
- Senior customers (60+) spend the most on average across all product categories.
- High Income customers (>\$75K) spend 4–5x more than Low Income customers.
- PhD and Master degree holders outspend Graduation-level customers.
- There is a strong positive correlation between Income and Total Spend.

3.5 Channel Analysis

- Store purchases dominate overall volume across all customer segments.
- High Spenders (top 10%) heavily favour Store and Catalog channels over web and deals.
- Young customers show relatively higher web purchase proportions compared to older segments.
- A significant segment of customers has high web visits (>5/month) but low actual spending — an under-served opportunity.
- Deals channel attracts predominantly Low Income, price-sensitive buyers who rarely respond to campaigns.

3.6 Complaint Analysis

- Only ~1% of customers have complained, indicating a generally satisfied base.
- Complainers have a lower average spend and response rate than non-complainers.
- Addressing complaints proactively could convert this small group into active responders.

4. Actionable Recommendations

1: Target High-Income, Graduate/PhD Customers First

This segment combines the highest spend per customer with the highest campaign response rates. Future campaigns should be specifically designed around their product preferences — Wine, Meat, and Gold — with personalised messaging tied to premium quality and exclusivity.

2: Invest in Store and Catalog Channels

High-value customers (top 10% spenders) use Store and Catalog channels far more than web or deals. Allocating more campaign budget to in-store promotions, physical catalogs, and loyalty programmes in these channels will directly reach the highest-spending customers.

3: Convert High-Visit, Low-Spend Web Customers

Customers who visit the website more than 5 times per month but have low spend represent a significant missed revenue opportunity. Implement targeted retargeting ads, abandoned-browse email flows, and exclusive web-only discounts to convert browsing intent into purchases.

4: Create Dedicated Senior Customer Campaigns

Senior customers (60+) are the highest-spending age group across nearly every product category. Campaigns tailored to their preferences, using channels they favour (Store, Catalog), with messaging focused on quality and value, are likely to generate strong ROI.

5: De-prioritise the Deals Channel for Premium Campaigns

The Deals channel attracts primarily low-income, price-sensitive customers who rarely respond to brand campaigns. Continued heavy investment in deals discounts margins without building long-term customer value. Reserve it for clearance or volume-driven objectives only.

6: Proactively Resolve Complaints

Although the complaint rate is only ~1%, complainers show measurably lower spend and response rates. A structured complaint resolution programme — including follow-up outreach and a recovery offer — could re-engage this group and improve their lifetime value.

5. Ideal Target Customer Profile

Based on the EDA and segment analysis, the profile of the most responsive and highest-value customer for future campaigns is:

Attribute	Ideal Value
Age Range	45–65 (Late Middle Aged to Senior)
Income Band	> \$75,000 (High)
Education	Graduation, Master, or PhD
Marital Status	Married
Children at Home	Not more than 1
Country	Spain, Australia, Canada
Preferred Channel	Store and Catalog
Top Products	Wine and Meat
Campaign History	Accepted more than 2 previous campaigns
Web Visits/Month	2–5 visits